Anthony Huang

https://magicalant.github.io/ | anthony.huang@student.isb.bj.edu.cn | MagicalAnt@outlook.com

Education

International School of Beijing — September 2023 – May 2027

Research & Internships

Independent Researcher, AI and Machine Learning in Computer Vision — April 2026 – May 2026

- Attended workshop led by UCLA Prof Ramin Ramezani, studied AI fundamentals and how to build, improve, and evaluate machine learning models;
- Individually completed a comparative robustness study between custom CNN and ResNet-18, demonstrated the fragility of CNNs in real-world applications and that clean benchmark accuracy can be misleading;
- Independent paper accepted by the 8th International Conference on Computing and Data Science (CDS_AST 2026).

Student Learner, "Diagnosing Alzheimer's Disease With AI" Science Camp — January 2026 – March 2026

- Collaborated with a medical AI mentor to use machine learning to diagnose Alzheimer's disease;
- Explored deep learning models, including LeNet, ResNet, UNet, ResUNet, and learned how to process MRI/CT images and develop diagnostic models;
- Earned a perfect final evaluation upon camp completion.

Core AI Team Intern, Airdoc (tech company focused on retinal imaging) — January 2025 – February 2026

- Built 3 custom models based on EfficientNet-V2-M with different loss functions (MAE, MSE, Cross-Entropy) to predict patient age and gender from their retinal images;
- Trained and compared model performance using Airdoc's proprietary retinal image dataset, offered loss-function selection guidance for the team.

Co-designer and Researcher of AI Navigation Cane — July 2024 – October 2024

- Volunteered at HongDanDan Visually Impaired Service Center, read books for the kids, learned about their common daily challenges, led them to visit Microsoft and experience AI technologies;
- Co-designed and implemented an AI navigation cane to improve mobility and independence for the visually impaired, using raspberry pi, CoCo model, webcam, ultrasonic sensors, etc.

Customer Engagement Intern, Airdoc — July 2024

- Received clinical training, practiced patient eye measurement on various instruments and devices in Optometry, Biometrics, Retinal Imaging, Tonometry, etc;
- Learned the causes and types of eye diseases and how to analyze and interpret patients' retinal images.

Game Developer — 2024

- Two-player PVP and PVE guessing dice game: developed in GoDot, animations, timers, etc;
- Roguelike PVE Ninja game: collaborated with graphic designers, developed in GoDot, with Scenes, Sprites, Collisionshape, sound, animation, background music, and multi-platform builds (macOS/Windows/Linux).

Leadership & Extracurricular Activities

President, Executive Member (9th Grade), ISB Chapter of Beijing Student Press — May 2024 – Present

- Led production of the club's flagship magazine semi-annually, 30-50 pages per issue;
- Expanded the club's focus from literary writing to visual arts, calligraphy, graphic design, and other works, launched regular school-life feature columns to better support the 2,000-student community;
- Organized team-bonding activities to foster a more engaging club atmosphere;
- Submissions increased by 600% since 2024.

President, ISB Chinese Debate Club — September 2025 – Present

- Organized regular club meetings, hosted mock debates, invited spectators to provide feedback;
- Recruited, trained, and led a team of 10 to participate in the 10th Zongheng Cup International School Mandarin Debate Tournament (104 teams in total), advanced to the quarterfinals;
- Earned Outstanding Individual Debater at the tournament.

Lead Playwright, Adaptation of *Thunderstorm,* for ISB Annual Assembly April 2026 – June 2026

- Key member of the 7-person playwright team to adapt the classic Chinese play "Thunderstorm" (雷雨), the second-ever Chinese production in ISB history, and the first to be performed by the 10th grade;
- Finished a 10,000+ word script, rewrote complex storylines to make it friendly for all grade levels and non-native audiences;
- Helped recruit key acting members for the play.

Playwright, Original Play *The Witch Hunters,* for ISB One Act Festival September 2025 – Present

- Individually developed a 20,000+ word fantasy-adventure script, "*The Witch Hunters,*" inspired by *Once Upon a time* and Fairy Tales such as *Hansel and Gretel*;
- Attended writing workshops hosted by ISB teacher Joseph Stewart and guest speakers to hone narrative skills;
- Helped recruit and coach actors for performance delivery, production scheduled for One Act Festival in late 2026.

Participant, Iowa Young Writers' Studio January 2025 – March 2025

- Took the 6-week online writing course "Get Dusty! Fiction Writing", learned how to vividly express my imagination by using the five senses to create more immersive and engaging narratives.

Keyboard, "Little Times" Band January 2024 – Present

- Co-founded the student band and performed at 2024 ISB's Annual Assembly;
- Met regularly and rehearsed with band members to adapt songs and prepare for live shows.

Photographer, ISB School Exhibit February 2026

- Attended the photography class in 10th grade, multiple works were selected for exhibit on campus;
- The street photography collection was voted the best by the class.

ISB High School Swimming Team September 2023 – Present

- Trained with the swim team, took part in friendly swim meets.

Honors & Awards

June 2026 Paper Accepted by The 8th International Conference on Computing and Data Science, Machine Learning and Neural Network Applications in Engineering (CDS_AST 2026), published in *Applied and Computational Engineering (Print ISSN: 2755-2721)*: *Robustness of Convolutional Neural Networks to Partial Test-Set Image Corruption on Fashion-MNIST*

April 2025 Outstanding Debater, Zongheng International School Mandarin Debate Tournament

2023-2024 The 4th Chinese Language Festival:

- Championship Awards in Modern Poem & Modern Essay
- Runner-up Awards in Ancient Poem Recital & Writing
- Award of Merit in Modern Poem Recital

July 2024 Beijing HongDanDan Visually Impaired Service Center – Honorary Volunteer Certificate

Skills & Hobbies

- Programming Languages: Java, Python, HTML, SQL;
- Skills: VSCode/Debugger, Data Structures, OOP, PyTorch, Tensorflow, CNN, ResNet, Database, 3D Modelling, Git
- Hobbies: Drum, Piano, Ultimate Frisbee, Swimming, Football, Ping-pong, Badminton.